

Rethinking models

Numerical agreement and time to draws

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Stanli runs Stan programs without a separate C++ compilation for each model. We evaluated the unchanged Stan programs and data generated by all 61 `ulam()` call sites in the second-edition code supplement. On the current build, 60 book call sites passed the numerical replay; m14.11 could not finish model preparation within five minutes. 58 book call sites completed all four timed seeds in both engines.

Including model compilation, the estimated time to a first fit was shorter with stanli for 56 of the 58 book call sites that completed in both engines. Much of this advantage comes from avoiding a separate C++ compilation. Full results include every incomplete case; completion and diagnostic flags must be considered alongside elapsed time.

Numerical agreement

At three fixed parameter vectors per fixture, we compared the log posterior, every derivative, and constrained/model-output values. Sixty-one of 62 fixtures passed, including the separate hurdle example: 32,316 values were compared, with a largest scaled discrepancy of $6.50\text{e-}14$. The tolerance was 10^{-9} . The remaining fixture, m14.11, timed out during preparation, before the comparison (issue #372). Pointwise agreement is not a proof of equality everywhere.

Examples: estimated time to a first fit

Seconds for one chain. CmdStan includes its recorded model-compilation stages; stanli preparation is already included. Screen S/C means stanli/CmdStan; review flags a diagnostic. See the appendix for all models and the four-seed ranges.

Book model / example	stanli	CmdStan	Screen S/C
m11.4 - Chimpanzee logistic regression	0.207	3.41	clear/clear
m12.4 - Ordered response, cutpoints only	0.672	89.3	clear/clear
m12.5 - Ordered response with predictors	283	165	clear/clear
m12.6 - Monotonic education effect	-	489	incomplete/clear
m13.4nc - Non-centered multilevel model	0.744	5.23	clear/clear
m14.8 - Gaussian-process model	0.839	5.53	clear/clear
m15.2 - Measurement-error model	0.083	3.57	clear/clear
m16.1 - Nonlinear height-weight model	3.60	11.6	clear/clear

Sampling quality matters

47 of the 61 book call sites completed and met the diagnostic screen in both engines. The others had diagnostic flags or did not complete. The book includes deliberately difficult models, so elapsed time alone is not evidence of reliable inference. We did not retune individual models to make diagnostics pass.

How to read the results

Each engine used four independent seeds, with 1,000 warmup iterations and 1,000 retained draws per seed. Timings below describe one chain, not a parallel four-chain fit. Each stanli CLI run includes Stan translation, model preparation, warmup, sampling, generated quantities and CSV output. CmdStan CLI times start from its compiled model; compilation is shown separately.

CSV precision follows the CLI defaults: eight significant digits for CmdStan and 17 for stanli. The numerical oracle uses separate high-precision values; sampling diagnostics use these retained CSVs.

Hardware and software

Apple M3 Ultra, 96 GiB RAM, macOS ARM64; Release build; one engine at a time. The toolchain was already installed. Rethinking 2.42, CmdStan 2.39.0 and posterior 1.7.0 were used. Runtime/compiler sources match main 2ae6c1d0; the manifest records all build identities. These are desktop measurements, not predicted laptop timings.

The scaled error is $\text{abs}(a-b) / \max(1, \text{abs}(a), \text{abs}(b))$. Log-posterior and derivative comparisons use the same unconstrained parameter values and include the transformation Jacobian.

Fixed settings and stopping rule

Seeds 1-4; target acceptance 0.8; maximum tree depth 10; random initialization. Every model used the same iteration budget. The book's model-specific tuning and chain-count choices were not reproduced.

For each seed, CmdStan ran first. Stanli was stopped at the smaller of three times that CmdStan CLI runtime and 900 seconds. CmdStan also had a 900-second ceiling. C++ compilation did not count toward the relative cap. This reference-first order was required by the cap and was not counterbalanced.

Incomplete cases

m12.6: stanli seed 1: timeout (900-second limit). m13.6: stanli seed 3: timeout (2.62-second limit). m14.11: no sampling comparison; preparation or the preceding numerical/timing gate did not finish. See the retained failure log.

Diagnostic screen

We checked the four runs as four chains. A clear screen requires no post-warmup divergences or maximum-tree-depth hits, finite R-hat no greater than 1.01, and bulk effective sample size of at least 400 for every nonconstant variable in the parameters block. Fixed correlation-matrix entries were verified and omitted. A flag calls for review; a clear screen is not proof that inference is reliable.

Under this screen, 48 of 58 complete stanli book fits and 49 of 60 CmdStan book fits were clear; 47 were clear in both. Several flagged cases are intentionally difficult teaching examples. Runtime is the cost of a fixed sampling budget, not time to a specified inferential accuracy.

Scope and sources

All 61 `ulam()` call sites in chapters 4-16 are included, representing 58 distinct Stan/data pairs. Repeated fits remain listed. The supplemental hurdle fixture is separate. `quap()`, direct `stan()` calls, exercises and lecture-only examples are outside this inventory.

Pinned second-edition code supplement (rethinking 2.42)
Stan: convergence and efficiency diagnostics
Preparation timeout (#372) | Ordered-regression timing (#373)

The accompanying CSV and JSON files contain every timing, diagnostic result and build/input identity. Run ID: fae5494c296cd547. Numerical replay compared 32,316 values across 61 fixtures, including model outputs. The m14.11 preparation failure is tracked in issue #372. Inputs were identical across engines; generated Stan programs were not rewritten.

Full timing appendix

Seconds: median [minimum-maximum] across four seeds. One chain has 1,000 warmup iterations and 1,000 retained draws. S/C = stanli/CmdStan diagnostic screen.

Book model	R code	stanli runtime	CmdStan runtime	CmdStan compile	Screen S/C
m9.1	9.14	0.030 [0.025-0.033]	0.053 [0.046-0.257]	3.09	clear/clear
m9.1	9.16	0.030 [0.026-0.032]	0.056 [0.046-0.254]	3.06	clear/clear
m9.2	9.22	0.018 [0.017-0.019]	0.025 [0.023-0.245]	2.50	review/review
m9.3	9.24	0.014 [0.013-0.014]	0.017 [0.017-0.233]	2.52	clear/clear
m9.4	9.26	0.970 [0.943-0.978]	1.41 [1.36-1.58]	2.55	review/review
m9.5	9.27	0.270 [0.264-0.280]	0.354 [0.323-0.563]	2.56	clear/clear
mp	9.28	0.022 [0.014-0.028]	0.067 [0.020-0.245]	2.29	clear/review
m5.8s	9.29	1.24 [1.22-1.26]	3.57 [3.48-3.73]	2.92	review/review
m5.8s2	9.30	0.873 [0.318-1.17]	2.59 [1.12-3.33]	2.86	review/review
m11.4	11.11	0.207 [0.206-0.218]	0.395 [0.387-0.612]	3.02	clear/clear
m11.5	11.19	0.320 [0.304-0.322]	0.653 [0.636-0.881]	3.09	clear/clear
m11.6	11.25	0.038 [0.035-0.039]	0.052 [0.049-0.254]	3.02	clear/clear
m11.7	11.29	0.014 [0.013-0.015]	0.018 [0.017-0.198]	2.81	clear/clear
m11.8	11.32	0.051 [0.048-0.055]	0.078 [0.071-0.272]	2.93	review/review
m11.9	11.45	0.012 [0.011-0.014]	0.015 [0.013-0.192]	2.60	clear/clear
m11.10	11.45	0.020 [0.020-0.021]	0.027 [0.026-0.221]	3.25	clear/clear
m11.11	11.49	0.141 [0.132-0.154]	0.219 [0.197-0.420]	3.54	review/clear
m_pois	11.61	0.012 [0.012-0.013]	0.014 [0.014-0.212]	2.51	clear/clear
m12.1	12.2	0.030 [0.029-0.031]	0.041 [0.037-0.246]	3.21	clear/clear
m12.2	12.6	0.107 [0.095-0.124]	0.159 [0.134-0.356]	3.97	clear/clear
m12.3	12.9	0.028 [0.027-0.030]	0.257 [0.236-0.446]	2.50	clear/clear
m12.3_alt	12.11	0.054 [0.053-0.061]	0.257 [0.227-0.426]	2.48	clear/clear
m12.4	12.16	0.672 [0.640-0.689]	86.1 [81.1-89.3]	3.19	clear/clear
m12.5	12.24	283 [273-291]	162 [159-166]	3.57	clear/clear
m12.6	12.34	capped (1/4)	485 [455-508]	4.24	incomplete/clear
m12.7	12.37	381 [310-389]	214 [191-228]	3.55	clear/clear
m13.1	13.2	0.052 [0.048-0.054]	0.073 [0.069-0.291]	2.96	clear/clear
m13.2	13.3	0.058 [0.056-0.061]	0.086 [0.080-0.298]	3.11	clear/clear
m13.3	13.13	0.069 [0.063-0.079]	0.080 [0.076-0.279]	3.04	clear/clear
m13.4	13.21	0.490 [0.415-0.622]	0.777 [0.675-1.13]	3.42	review/review
m13.5	13.23	0.243 [0.237-0.248]	0.438 [0.414-0.631]	3.24	clear/clear

To estimate a first fit, add CmdStan compilation to its runtime; stanli preparation is already included. Compilation was measured once, including Stan-to-C++ translation and the C++ build. First-fit totals are sums of measured stages. Four-seed ranges describe run-to-run variation, not confidence intervals.

Full timing appendix (continued)

Seconds: median [minimum-maximum] across four seeds. One chain has 1,000 warmup iterations and 1,000 retained draws. S/C = stanli/CmdStan diagnostic screen.

Book model	R code	stanli runtime	CmdStan runtime	CmdStan compile	Screen S/C
m13.6	13.25	capped (1/4)	1.02 [0.872-1.16]	3.40	incomplete/review
m13.7	13.26	0.016 [0.014-0.018]	0.022 [0.020-0.219]	2.30	review/review
m13.7nc	13.27	0.012 [0.011-0.013]	0.014 [0.014-0.211]	2.34	clear/clear
m13.4b	13.28	0.406 [0.339-0.546]	0.607 [0.513-0.958]	3.31	review/review
m13.4nc	13.29	0.744 [0.724-0.790]	1.90 [1.84-2.11]	3.33	clear/clear
m14.1	14.12	0.314 [0.305-0.318]	0.403 [0.391-0.665]	6.92	clear/clear
m14.2	14.18	2.61 [2.22-3.12]	3.74 [3.46-4.49]	7.08	review/review
m14.3	14.19	0.993 [0.970-1.03]	1.74 [1.72-1.97]	5.92	clear/clear
m14.4	14.24	0.031 [0.029-0.035]	0.079 [0.072-0.278]	2.81	clear/clear
m14.5	14.25	0.052 [0.046-0.053]	0.166 [0.150-0.351]	2.82	clear/clear
m14.6	14.26	2.73 [2.67-2.90]	4.04 [3.81-4.11]	6.67	clear/clear
m14.4x	14.27	0.032 [0.030-0.034]	0.077 [0.075-0.264]	2.79	clear/clear
m14.6x	14.27	2.77 [2.64-2.88]	4.01 [3.85-4.10]	6.67	clear/clear
m14.7	14.31	4.60 [4.51-4.69]	6.55 [6.47-6.62]	8.86	clear/clear
m14.8	14.39	0.839 [0.815-0.901]	1.05 [0.947-1.06]	4.48	clear/clear
m14.8nc	14.46	0.807 [0.732-1.00]	1.19 [1.11-1.58]	5.72	clear/clear
m14.9	14.49	6.21 [6.09-6.82]	6.14 [5.80-6.56]	4.06	clear/clear
m14.10	14.51	6.14 [5.66-6.29]	6.17 [6.07-6.30]	4.06	clear/clear
m14.11	14.52	not measured	not measured	-	incomplete/incomplete
m15.1	15.3	0.049 [0.048-0.054]	0.103 [0.093-0.315]	3.20	clear/clear
m15.2	15.5	0.083 [0.074-0.084]	0.140 [0.131-0.359]	3.43	clear/clear
m15.3	15.12	0.240 [0.231-0.245]	0.305 [0.296-0.523]	2.68	clear/clear
m15.4	15.13	0.206 [0.193-0.222]	0.259 [0.238-0.456]	2.64	clear/clear
m15.5	15.17	0.037 [0.033-0.041]	0.064 [0.060-0.266]	3.23	clear/clear
m15.6	15.19	0.022 [0.022-0.024]	0.033 [0.030-0.213]	2.97	clear/clear
m15.7	15.22	0.586 [0.547-0.648]	0.553 [0.533-0.759]	7.18	clear/clear
m15.8	15.30	0.052 [0.051-0.056]	0.113 [0.103-0.310]	2.83	clear/clear
m15.9	15.31	0.072 [0.072-0.078]	0.158 [0.149-0.342]	3.00	clear/clear
m16.1	16.2	3.60 [3.56-3.66]	8.58 [8.49-8.70]	3.02	clear/clear
m16.4	16.11	0.081 [0.070-0.088]	0.235 [0.219-0.413]	2.90	clear/clear

To estimate a first fit, add CmdStan compilation to its runtime; stanli preparation is already included. Compilation was measured once, including Stan-to-C++ translation and the C++ build. First-fit totals are sums of measured stages. Four-seed ranges describe run-to-run variation, not confidence intervals.

Supplemental fixture (not a book model)

Hurdle Poisson: stanli 0.014 [0.014-0.017] s; CmdStan 0.018 [0.018-0.221] s; CmdStan compilation 2.59 s. Screen: clear/clear.